Veer Narmad South Gujarat University,
Surat.

Minor Project Report

On

"AI BASED COURSE GENERATOR"

As a partial fulfillment for degree of
Bachelor Of Computer Application
T.Y.B.C.A.(5th Semester)
Year 2025-2026

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No.NCSC/

Certificate

This is to certify that **Mr. Krish Dalsaniya** (Exam No. 1214) of **Narmada College of Science and Commerce, Bharuch**, has successfully completed his **Minor Project Report** on *"AI-Based Course Generator"* in partial fulfillment of the requirements for the degree of **Bachelor of Computer Applications (B.C.A.)** from **Veer Narmad South Gujarat University** for the academic year **2025–26**.

He carried out the project under the guidance of **Mr. Chetan Parmar**. The said report is based on his bona fide work done by him.

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Internal Guide

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H.O.D., B.C.A. Programme,
NCSC, Zadeshwar, Bharuch

ACKNOWLEDGEMENT

I am delighted to present this report on the *AI-Based Course Generator* project.

I would like to express my deepest appreciation to all individuals who contributed to its success, particularly my project guide, **Mr. Chetan Parmar**, for his constant guidance and support throughout the project.

I also acknowledge the valuable encouragement and assistance received from the **Computer Department staff** and my peers.

I am sincerely grateful to **Narmada College of Science & Commerce, Bharuch**, for providing a conducive learning environment.

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THANKING YOU

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Project Profile

Project Title	-	**AI Course Generator**
Project Description	-	AI Course Generator is an intelligent Android application that empowers educators to create full-fledged courses using AI. Users input a course title and difficulty level, and the app leverages Google Gemini and Groq, Cerebras to generate structured outlines, detailed content and video suggestions in real time. Learners can generate personalized course on any topic.The app uses Server-Sent Events (SSE) for live content streaming and supports multi-provider AI switching for optimal performance.
Front-end	-	Kotlin, Jetpack Compose, Material 3
Back-end	-	Express js, Postgres Express JS, Node JS (docker)
AI Engine	-	Google Gemini 2.5 , Groq, Cerebras and Server-Sent Events (SSE)
IDE	-	Android Studio
Hardware Interface	-	Processor: 11th Gen Intel(R) Core(TM) i5-11400H RAM: 16GB HDD/SSD: 512 GB Operating System : Windows 11 HOME (64 bit) Laptop
Application Software		Android Studio (Koala // 2024.1.1)

Project Overview

AI Course Generator App is an intuitive Android application designed to help users easily create complete course structures and content using advanced AI providers.
The app provides a simple interface where users can input course details, generate outlines, edit topics, and generate detailed content for each subtopic using their preferred AI model.
Administrators can manage the entire course database, users, and available AI providers.

Key Features:

1. Secure Login and Registration (JWT-based)
2. AI-Generated Course Outlines
3. Editable Outline Structure (add/remove subtopics)
4. Batch Content Generation for Each Subtopic
5. Multiple AI Providers (Groq, Cerebras, Gemini)
6. Admin Management Panel (courses, providers)
7. Encrypted Data Storage using EncryptedSharedPreferences

User Pages

1. **Login Page** – Allows users to securely log into the application using JWT authentication.
2. **Registration Page** – Enables new users to register with basic credentials.
3. **Course Input Page** – User provides course details such as:
    - Course Name
    - Description
    - Number of Units
    - Difficulty Level
    - Include YouTube Videos (toggle)
    - Preferred AI Provider (Groq, Cerebras, Gemini)
4. **Outline Generation Page** – After input, user taps “Generate Outline” to fetch an AI-generated outline based on their input.
5. **Outline Editor Page** – Users can modify the AI-generated outline by adding, removing, or renaming subtopics.
6. **Save Outline Page** – Allows users to save their finalized outline to the database.
7. **Content Generation Page** – Generates full course content in batches for each subtopic and stores it in the database.
8. **My Courses Page** – Displays a list of all generated courses by the user, with options to view or delete them.

Admin Pages

1. **Admin Login Page** – Allows admin to securely log in using separate credentials.
2. **Dashboard Page** – Displays overview of total users, courses, and active AI providers.
3. **Course Management Page** – Enables admin to view, delete, or manage all user-generated courses.

4. **Provider Management Page** – Admin can enable or remove AI providers (Groq, Cerebras, Gemini) from being available in the app.
5. **Course Generation Page** – Admin can generate courses manually using the same course input and generation flow as users (for demo or test purposes).

TOOLS USED

FRONT END

What is an Android Studio?

Android Studio, an Integrated Development Environment (IDE), serves as a comprehensive workshop for building Android apps.

This powerful toolset provides real-time design previews, enabling you to visualize your app's layout and user interface. Additionally, Android Studio's advanced code editor detects errors in your code, offering corrective suggestions to streamline the development process.

How Does An Android Studio Work?

Android Studio empowers developers to craft their ideal apps through efficient coding and seamless project management. Its essential features enable effortless code refinement, optimization, and debugging, ensuring a seamless user experience. With Android Studio's simulation capabilities, you can preview your app's performance on mobile or computer devices in real-time.

The Integrated Development Environment (IDE) streamlines the development process by identifying and resolving issues promptly. By consolidating multiple developer tools into one platform, Android Studio significantly simplifies and accelerates app development, saving valuable time and resources.

The Key Elements of Android Studio

Android Studio comprises four core elements that work in harmony to facilitate Android app development. Understanding these elements is crucial for effective app creation:

1. Software Development Kits (SDK): This element encompasses essential tools, documentation, and frameworks necessary for development. The SDK also includes an emulator for testing and tracking progress.

2. Integrated Development Environment (IDE): The IDE serves as the central hub for writing and managing code. It integrates the SDK components, plugins, and other tools, streamlining the development process.

3. Programming Languages: Android Studio supports two official languages, Java and Kotlin, and also allows the use of C and C++ through third-party tools. However, compatibility issues may arise with certain third-party apps.

4. Libraries: These libraries enable rapid code writing through automated systems, error detection, and rewriting. The API system recognizes coding patterns, offering suggestions and enhancing the development experience.

These four elements form the foundation of Android Studio, working together to provide a comprehensive and efficient development environment for creating Android apps.

Android Studio - Advantages

Android Studio is a tool that helps you as a developer to build an android app with absolute proficiency. It holds several advantages for you, which help you create high-quality applications for any android device.

Let's have a quick look over the Android Studio's Advantages.

1. Intelligent Code Editor

Android Studio has an intelligent Code editing system that helps you code faster. This system can easily detect any code errors and mark them and provide suggestions for changes.

It helps you to be more efficient with automated code completion. It reads the previous code writing patterns and provides completion codes according to the previously written codes.

2. Fast & Feature-rich Emulator

Android Studio features a built-in Android Emulator, enabling developers to test and refine their apps on a virtual Android device. This emulator allows you to run and interact with your app directly on your computer, simulating a real Android environment.

3. Code Templates & Lint checks

Android Studio's Code Templates feature is a valuable asset for developers of all levels, providing pre-built code patterns that simplify the writing of complex code. This feature serves as a guiding system, streamlining the development process.

Additionally, Android Studio's Lint Check feature ensures the structural integrity of your code, identifying weaknesses and providing suggestions for improvement. This feature helps maintain high-quality code, saving time and resources by: Detecting structural flaws ,Offering corrective suggestions ,Enhancing code maintainability

By leveraging Code Templates and Lint Check, developers can work more efficiently, producing robust and well-structured code with ease

4. Testing Tools and Frameworks

Android Studio Test your app's code frameworks with Android Studio's JUnit4 and Functional-UI tools. They record code, generate tests, and simulate UI interactions to ensure seamless app performance

5. Flexible System Build

Android Studio's UI system is designed for maximum flexibility, allowing developers to customize their workspace, integrate third-party tools and services, and configure their environment to optimize productivity and workflow.

6. C++ & NDK Support

Android Studio supports C and C++ development, enabling easy editing of C++ projects and creation of JNI components, with syntax highlighting and debugging tools for efficient coding.

7. Eliminates Tiresome Tasks

The GUI tool in Android Studio reduces development time by automating routine tasks, including layout design, APK analysis, and translation.

Android Studio - disadvantages

No app is completely flawless, and Android Studio is no exception, balancing benefits with drawbacks.

1. High System Requirements

Android Studio's minimum system requirements, including an i3 processor and 4GB of RAM, can be a significant obstacle for users with lower-end computers, limiting its accessibility.

2. System Lags

The app's system lag is a sudden cause of events, which might not occur. Sometimes when you're writing code on Android Studio, its system freezes and stops working and sometimes restarts on its own. Thus, affecting the overall efficiency of the work.

3. Slow Emulator

The emulator in Android Studio Provides is slow compared to other third-party android emulators. This slow emulator interface can't keep up with more than one app testing.

4. Heavy Ram Usage

Android Studio is a RAM-hungry app that needs at least 4 gigs of ram to run properly. If your computer has a lower RAM capacity, then you won't be able to get the best experience with this app.

5. Long Installation Process

The long installation process is another drawback of the Android Studio. It has a lengthy installation system that requires at least 3 minutes to complete, no matter your computer's configuration.

Frontend

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1. Kotlin + Jetpack Compose

Jetpack Compose is a modern declarative UI toolkit for building native Android applications using Kotlin.
It eliminates the need for XML layouts and allows developers to design responsive UIs through composable functions.
In this project, Jetpack Compose is used to build all user interfaces — including login, registration, home, course generation, and admin dashboards.

Features

- Declarative UI updates with state management
- Navigation component for seamless screen transitions
- Material Design 3 support for a consistent UI
- Integration with ViewModel and LiveData (MVVM architecture)

Advantages

- Reduces boilerplate code
- Real-time UI preview and hot reload
- Easier integration with backend APIs
- Improved performance over XML-based layouts

Disadvantages

- Requires learning new declarative patterns
- Limited backward compatibility for older Android versions

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2. Kotlin (Frontend Logic)

Kotlin is used to handle frontend business logic and connect the Jetpack Compose UI with backend APIs.
It manages user inputs, API calls (using Retrofit), and data storage (using Room or SharedPreferences for local caching).

Features

- Null safety and concise syntax
- Coroutines for asynchronous API calls
- Type-safe HTTP request handling via Retrofit

Advantages

- Cleaner and safer than Java
- Strong community support
- Great interoperability with existing Android libraries

Disadvantages

- Slower compilation at times compared to Java

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3. Retrofit

Retrofit is a type-safe HTTP client used to connect the frontend with the backend Node.js APIs.
It handles authentication, course generation requests, and data retrieval for both users and admins.

Features

- Supports JSON serialization/deserialization using Gson
- Easy integration with JWT for secure communication
- Built-in support for headers and query parameters

Advantages

- Reduces boilerplate network code

- Handles background threading with coroutines

Disadvantages

- Requires explicit error handling for API failures

Backend

4. Node.js + Express.js

Node.js provides the server runtime environment for backend logic, while Express.js acts as the web framework for handling HTTP requests, routes, and middleware.
The backend is responsible for authentication, role-based access control (RBAC), course generation logic, and AI API communication.

Features

- RESTful API endpoints for frontend integration
- Middleware-based JWT authentication system
- Role-based route protection for Admin/User
- AI provider integration (Groq, Cerebras, Gemini)

Advantages

- Asynchronous and fast I/O operations
- Easy to scale for multiple concurrent users
- Large ecosystem of NPM libraries

Disadvantages

- Requires careful error handling for async operations
- Single-threaded — CPU-heavy tasks must be offloaded

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5. PostgreSQL (via Aiven Cloud)

PostgreSQL is the primary relational database used for structured data storage.
It stores user accounts, course details, AI provider configurations, and generated content.
Aiven Cloud hosting ensures reliability and automatic backup support.

Features

- Tables for users, courses, subtopics, and AI providers

- Foreign key constraints for relational mapping
- Indexing for optimized query performance
- SSL-secured remote connections

Advantages

- Open-source and highly reliable
- Supports both relational and JSONB data types
- Integrates seamlessly with Node.js using `pg` library

Disadvantages

- Aiven's free tier has resource limits
- More complex setup than NoSQL databases

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6. JWT (Authentication System)

JSON Web Tokens are used to maintain a stateless authentication mechanism.
After login, the server issues a signed token that the frontend includes in all subsequent API requests.

Features

- Access and Refresh token pair
- Role-based token claims (Admin/User)
- Middleware for verification and expiry checks

Advantages

- Secure and stateless
- Ideal for mobile communication (no cookies)

Disadvantages

- Token revocation must be handled manually

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7. Role-Based Access Control (RBAC)

RBAC defines different access levels in the app to protect sensitive routes.
Roles include:

- **User**: Can generate courses using available AI providers
- **Admin**: Can manage providers, monitor system activity, and manage users

Advantages

- Enforces security and access rules

- Simplifies permission management

Disadvantages

- Requires consistent middleware configuration

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8. AI Providers (Groq, Cerebras, Gemini)

The system integrates multiple AI providers through API calls.
Users can select which provider to use when generating course outlines and content.

Features

- Dynamic API selection during course creation
- Auto-fetching of topic and subtopic content
- Rate limiting and error handling for each provider

Advantages

- Provides flexible and adaptive AI content generation
- Improves reliability by offering multiple fallback options

Disadvantages

- Requires active internet connection
- May incur usage limits or API costs

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9. Thunder Client (API Testing)

Thunder Client (VS Code extension) was used to test all backend routes, authentication flows, and provider integrations during development.

Features

- Token-based testing
- Environment variables for dev/prod configs
- Response preview and request history

Advantages

- Lightweight and fast alternative to Postman
- Perfect for small-to-medium projects

Disadvantages

- No scripting or automated test suites

Goal of Project

"AI-Powered Personalized Course Generator" is an innovative learning platform designed to revolutionize how people learn online. In an era where vast information is scattered across the internet and AI tools require precise prompting to yield valuable responses, many learners struggle to stay consistent, structured, and motivated. This project addresses that challenge by enabling users to generate complete, personalized, and structured learning paths using advanced AI models — **Groq**, **Cerebras**, and **Gemini** — all integrated within a single unified application.

The app's mission is to eliminate the confusion of "what to learn" and "how to learn" by automating the entire course creation process. Through a simple interface, users can input key details — such as course name, description, number of units, difficulty level, and media preferences — and the system intelligently generates a well-organized outline and detailed content, tailored to the user's exact goals.

This project blends the power of **multi-model AI**, **personalization**, and **user-driven flexibility**, creating a truly next-generation educational experience. By doing so, it aims to make **structured, self-paced learning accessible to anyone** — without needing expert knowledge of prompt engineering or advanced AI skills.

Objective of Project

1. **Simplify AI-Based Learning:**
   Allow users to create customized and well-structured course content without needing technical knowledge of AI or prompt engineering.
2. **Combat Procrastination through Structure:**
   Help learners overcome procrastination by providing a guided, goal-oriented, and personalized learning roadmap generated by AI.
3. **Multi-Model AI Integration:**
   Integrate leading AI models — **Groq**, **Cerebras**, and **Gemini** — within a single application, giving users diverse and flexible course generation options.
4. **User-Friendly Course Customization:**
   Empower users to edit and refine AI-generated outlines by adding or removing subtopics before saving, ensuring total personalization.
5. **Role-Based Access Control (RBAC):**
   Maintain a secure and scalable environment by implementing role-specific access for users and administrators.
6. **Automated Course Content Generation:**
   Generate detailed subtopic content in batches, allowing seamless, high-speed content creation and saving directly to the database.
7. **Smart Admin Dashboard:**
   Provide admins with full control over the platform — including course management, AI provider configuration, and user access oversight.
8. **Scalable Backend and Secure Storage:**
   Utilize **Node.js + Express.js** for backend and **Aiven PostgreSQL** for cloud database, ensuring secure data handling, smooth API communication, and scalability.
9. **Modern and Adaptive Frontend:**
   Implement **Jetpack Compose** for a modern Android UI that enhances user interaction, responsiveness, and usability.
10. **Quality Assurance and Testing:**
    Perform backend testing using **Thunder Client**, ensuring all endpoints, authentication flows, and database operations are reliable and efficient.

The **AI-Powered Personalized Course Generator** aims to redefine the way individuals learn and design their educational journeys. Traditional online learning platforms either offer rigid pre-designed courses or rely on learners to manually search and compile fragmented information. Even with modern AI tools, users often struggle to get meaningful educational outputs — since success depends heavily on their ability to craft effective prompts.

This project eliminates that barrier by combining **personalization, automation, and intelligence**.
It allows anyone from students to professionals to generate an end-to-end course on any topic they desire, built by advanced AI models that understand context, complexity, and learning goals.

The scope extends beyond standard course generation. It builds a **foundation for AI-driven learning ecosystems**, where:

- Users can create learning plans aligned with personal goals or skill gaps.
- Educators and organizations can rapidly generate educational material.
- Multiple AI providers can coexist and compete within one application, ensuring the highest-quality output.

The platform ultimately bridges the gap between **human curiosity** and **AI capability**, turning open-ended knowledge into structured, actionable learning one personalized course at a time.

Functions Performed

1. User:

A **User** is anyone who wants to generate a personalized course using AI models. Users can input course details, generate structured outlines, edit them, and produce full content for each subtopic.

Functions

- User can **Register** to create an account.
- User can **Login** securely using JWT-based authentication.
- User can **Input course details** such as course name, description, number of units, difficulty level, and video inclusion preference.
- User can **Select AI Provider** (Groq, Cerebras, or Gemini) for generating outlines or content.
- User can **Generate Course Outline** automatically using AI.
- User can **Edit Course Outline** by adding or removing subtopics before saving.
- User can **Save Course Outline** to the database.
- User can **Generate Course Content** for each subtopic in batches.
- User can **Logout** securely from the system.

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2. Admin:

An **Admin** is responsible for managing the entire platform, including courses, AI providers, and overall system configuration. The Admin also has the ability to create, manage, and delete courses.

Functions

- Admin can **Register/Login** with elevated privileges.
- Admin can **View, Delete Courses** created by users or the system.
- Admin can **Generate Courses** manually using the same course creation process.
- Admin can **Manage AI Providers** (Groq, Cerebras, Gemini) — add or remove them as available models.
- Admin can **Manage Role-Based Access Control (RBAC)** for system security.
- Admin can **Delete Users or Courses** violating terms or for maintenance.

Module Specification

a) User Module

Course Generation Management

- Input course details (title, description, difficulty, etc.).
- Select preferred AI provider for outline generation.
- Generate course outline using selected model.
- Save finalized course structure to the database.

Content Generation Management

- Generate course content for each subtopic automatically.
- Fetch and display generated content in a structured format.
- Support batch generation for large course outlines.

Personalization Settings

- Choose AI provider (Groq / Cerebras / Gemini) per course.
- Toggle inclusion of YouTube video suggestions.

b) Admin Module

AI Provider Management

- Add or remove integrated AI models (Groq, Cerebras, Gemini).
- Configure provider-specific API settings.
- Monitor usage metrics per provider.

Course Management

- View all courses generated by all users.
- Delete courses when required.

Database Design

Users table :-

Column name	Data Type	Description
id	UUID	Primary key
email	VARCHAR	
pass_hash	Text	
role	VARCHAR	
createdAt	DateTime	

- Courses Table:

Column name	Data Type	Description
id	UUID	Primary key
created_by	UUID	Foreign key referencing users(id)
title	Text	Course title
description	Text	Course description (optional)
difficulty	Text	Difficulty level ('Beginner', 'Intermediate', 'Advanced')
include_videos	Boolean	Whether the course includes videos (default FALSE)
status	Text	Course status ('draft', 'generating', 'ready')
outline_json	JSONB	JSON format outline for the course
outline_generated_at	TIMESTAMP	Timestamp when the outline was generated (optional)
content_generated_at	TIMESTAMP	Timestamp when the content was generated (optional)
is_public	Boolean	Whether the course is public (default FALSE)
created_at	TIMESTAMP	Course creation timestamp, default to NOW()

Subtopics Table:-

Column name	Data Type	Description
Id	UUID	Primary key
course_id	UUID	Foreign key referencing units(id)
title	Text	Subtopic title
content	Text	Content for the subtopic (optional)
position	Int	Position/order of the subtopic within the unit

Videos Table :-

Column name	Data Type	Description
id	UUID	Primary key
subtopic_id	UUID	Foreign key referencing subtopics(id)
title	Text	Video Title
youtube_url	Text	YouTube video URL (unique) Video thumbnail URL (unique)
thumbnail	Text	Video thumbnail URL (unique)
duration_second	INT	Video duration in seconds

User Courses Table:-

Column name	Data Type	Description
id	UUID	primary key
user_ id	UUID	Foreign key referencing users(id)
course_id	UUID	Foreign key referencing courses(id)
joined_at	TIMESTAMP	Timestamp when the user joined the course

Refresh_Token Table:-

Column name	Data Type	Description
id	UUID	primary key
user_ id	UUID	Foreign key referencing users(id)
token	Text	Refresh token
ip_address	Text	IP address of the user
expires_at	TIMESTAMP	Expiry timestamp of the refresh token
created_at	TIMESTAMP	Timestamp when the refresh token was created

Units Table:-

Column name	Data Type	Description
course_id	UUID	Foreign key referencing courses(id)
title	VARCHAR	
position	INT	
id	INT	PK

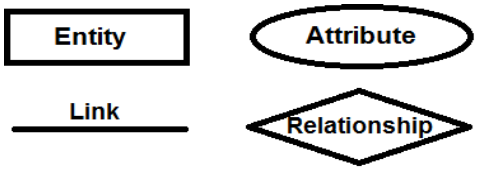
ER Diagram

E-R (Entity-Relationship) Diagram is used to represents the relationship between entities in the table.

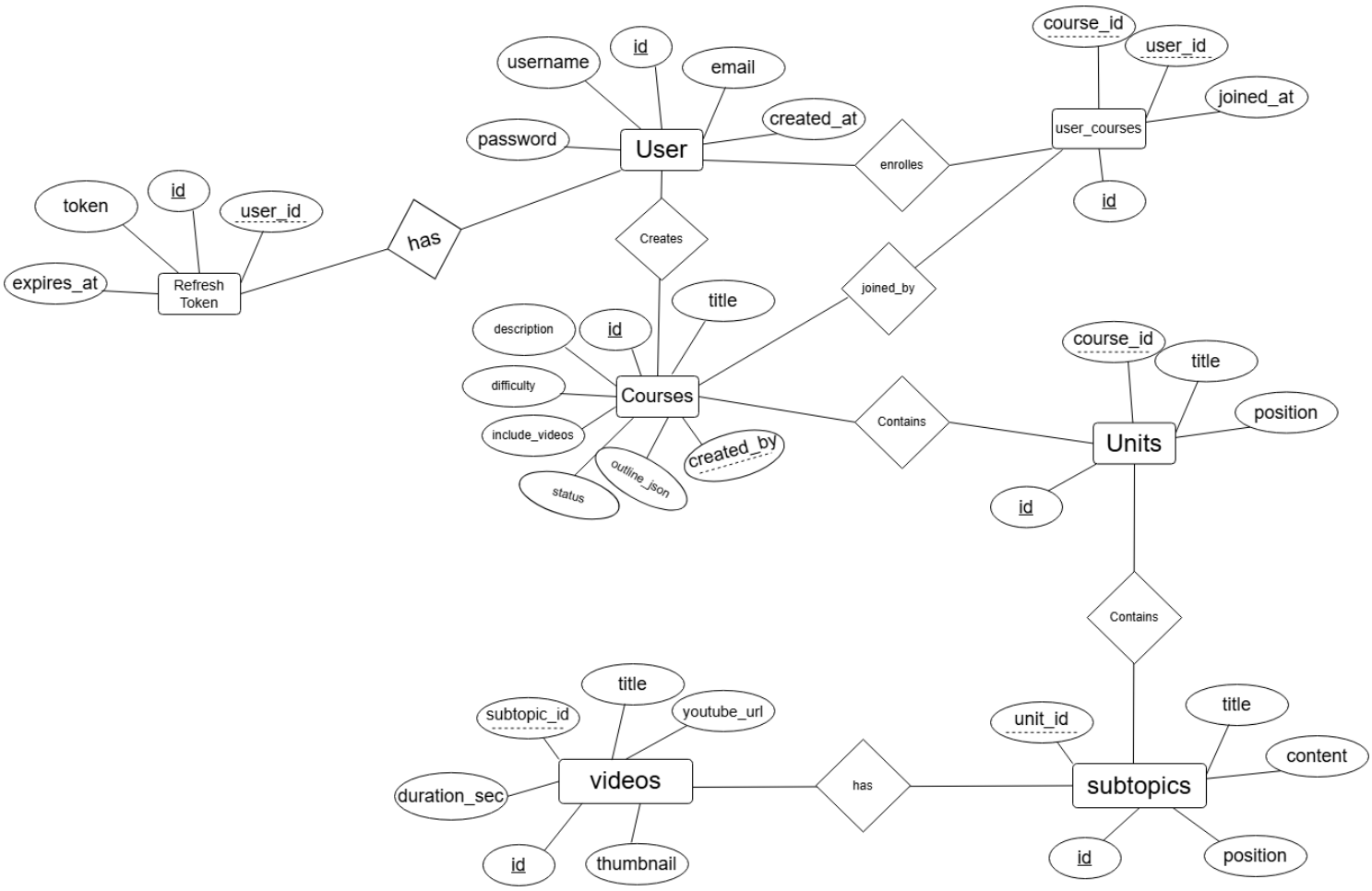
ER Notation

There is no standard for representing data objects in ER diagrams. The symbols used for the basic ER constructs are:

- **Entities** are represented by labeled rectangles. The label is the name of the entity. Entity names should be singular nouns.
- **Relationships** are represented by a solid line connecting two entities. The name of the relationship is written above the line. Relationship names should be verbs
- **Attributes**, when included, are listed inside the entity rectangle. Attributes which are identifiers are underlined. Attribute names should be singular nouns.
- **Links** are represented by a solid line connecting attributes to entity or entity to relationship



ER Diagram - General Symbols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expires_at
has
Courses
description
id
title
difficulty
include_videos
status
outline_json
created_by
Contains
Units
course_id
title
position
id
Contains
subtopics
unit_id
title
content
id
position
has
videos
title
subtopic_id
youtube_url
duration_sec
id
thumbnail

Data Flow Diagram

A data flow diagram is graphical tool used to describe and analyze movement of data through a system. These are the central tool and the basis from which the other components are developed. The transformation of data from input to output through processed, may be described logically and independently of physical components associated with the system. These are known as the logical data flow diagrams. The physical data flow diagrams show the actual implements and movement of data between people, departments and workstations. A full description of a system actually consists of a set of data flow diagrams. Using two familiar notations Yourdon, Gane and Sarson notation develops the data flow diagrams. Each component in a DFD is labeled with a descriptive name. Process is further identified with a number that will be used for identification purpose. The development of DFD'S is done in several levels. Each process in lower-level diagrams can be broken down into a more detailed DFD in the next level. The top-level diagram is often called a "*context diagram"*.

A DFD is also known as a "bubble chart" has the purpose of clarifying system requirements and identifying major transformations that will become programs in system design. So it is the starting point of the design to the lowest level of detail. A DFD consists of a series of bubbles joined by data flows in the system

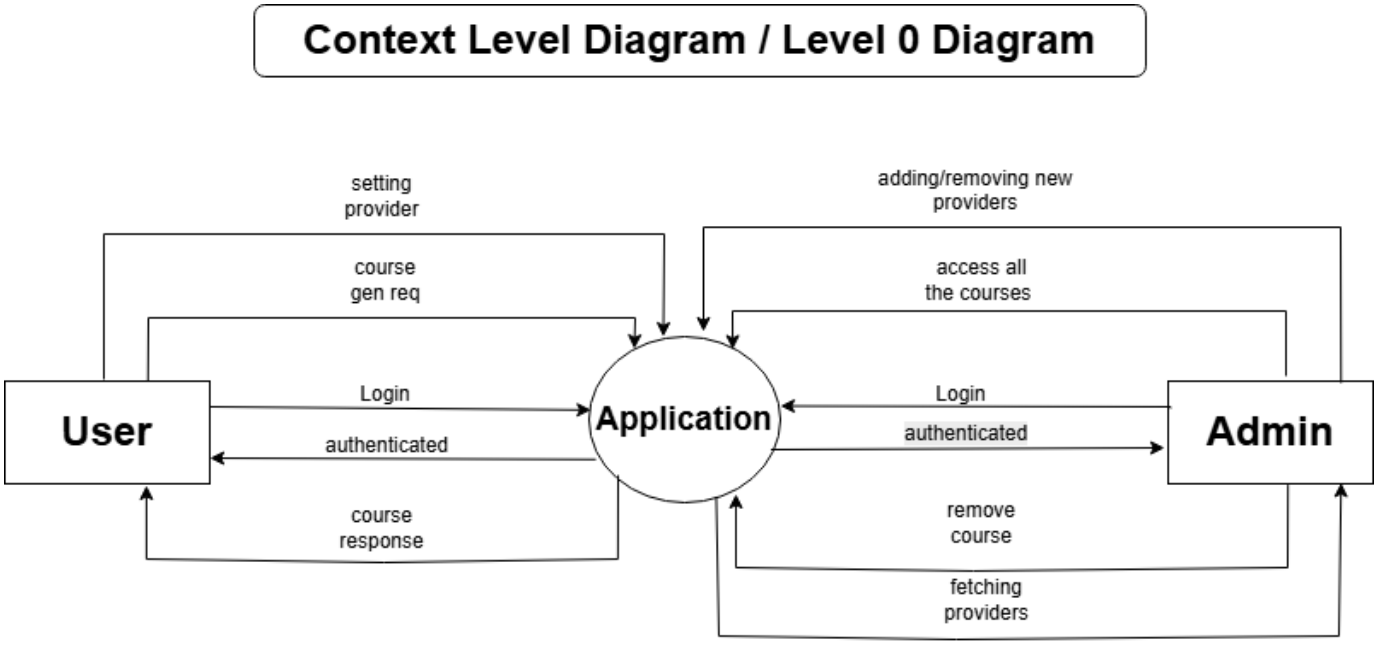
In the DFD, there are four symbols

1. A square defines a source(originator) or destination of system data
2. An arrow identifies data flow. It is the pipeline through which the information flows. Data move in a specific direction from an origin to a destination.
3. A circle or a bubble represents a process that transforms incoming data flow into outgoing data flows.
4. An open rectangle is a data store, data at rest or a temporary repository of data

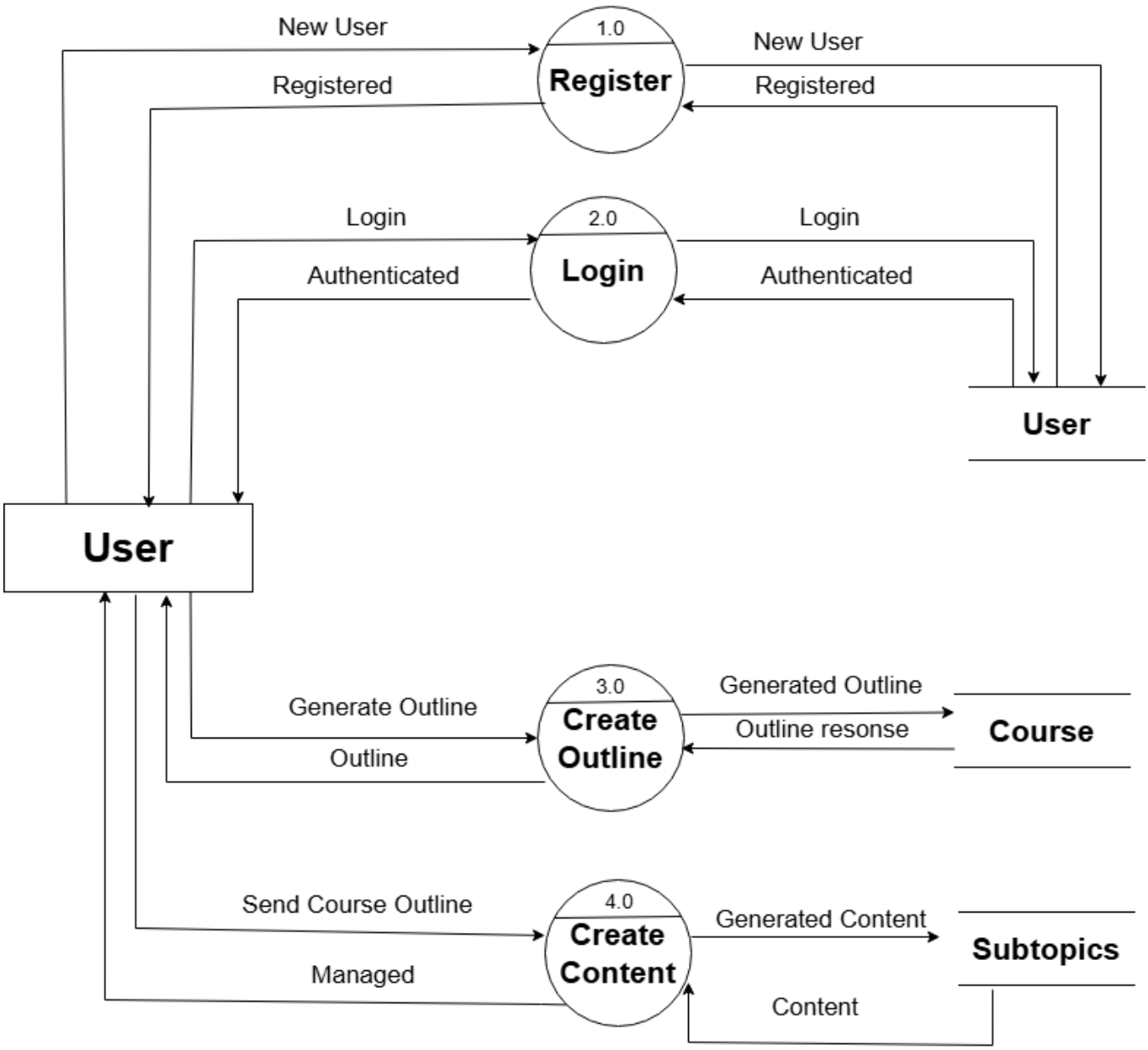
Symbols	Description
	Process that transforms data flow
	Data Flow
	Source or Destination of data (Entity)
	Data Store

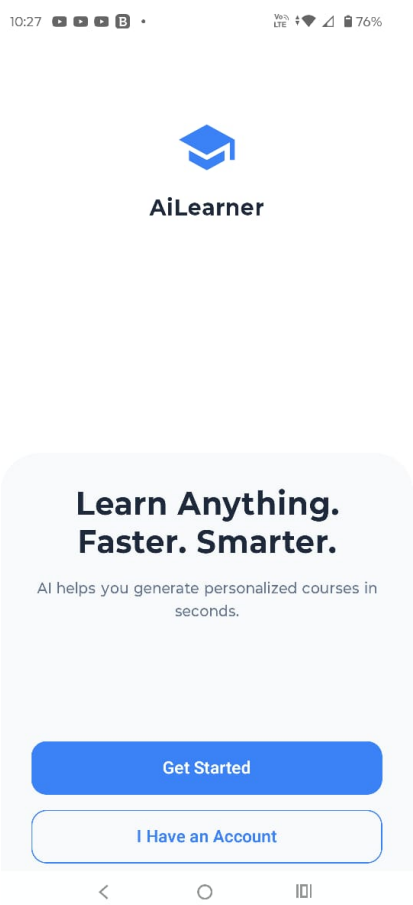
Context Level Diagram (Level 0)

It contains a single process, but it plays a very important role in studying the current system. The context diagram defines the system that will be studied in the sense that it determines the boundaries. Anything that is not inside the process identified in the context diagram will not be part of the system study. It represents the entire software element as a single bubble with input and output data indicated by incoming and outgoing arrows respectively.



Level 1 Diagram

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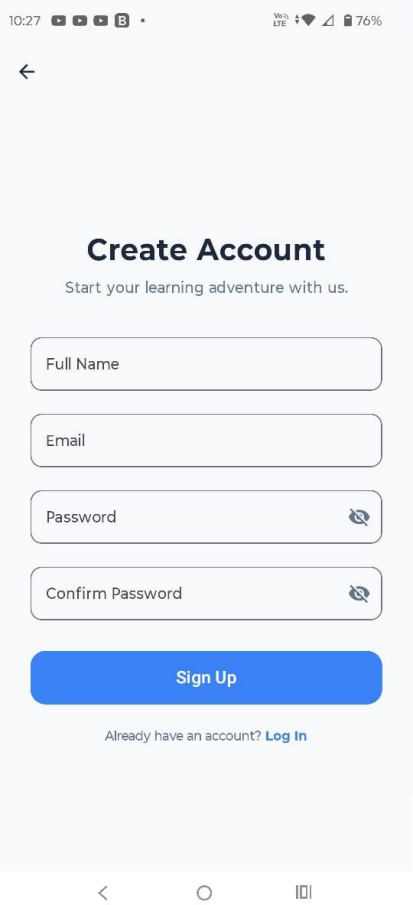
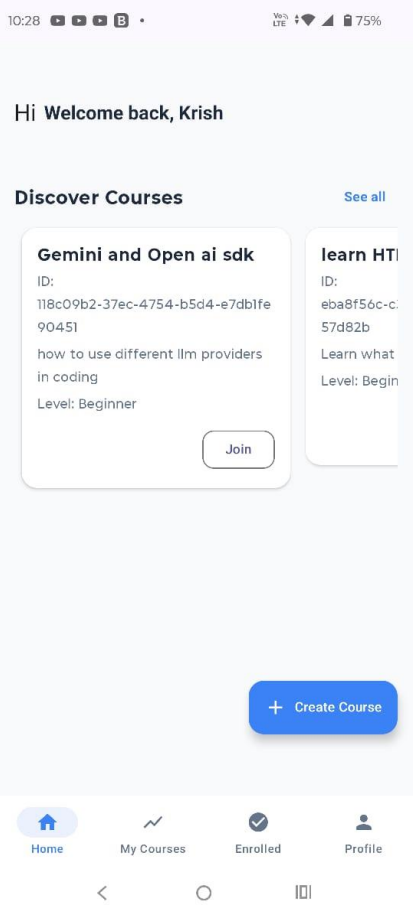
AiLearner
Learn Anything.
Faster. Smarter.
AI helps you generate personalized courses in seconds.
Get Started
I Have an Account


10:27
76%
Create Account
Start your learning adventure with us.
Full Name
Email
Password
Confirm Password
Sign Up
Already have an account? Log In


10:28
75%
Hi Welcome back, Krish
Discover Courses
See all
Gemini and Open ai sdk
ID:
118c09b2-37ec-4754-b5d4-e7db1fe90451
how to use different llm providers in coding
Level: Beginner
Join
learn HT
ID:
eba8f56c-c
57d82b
Learn what
Level: Begin
Create Course
Home
My Courses
Enrolled
Profile

10:28 75%

My Courses

demo course
Course ID: bdb6d3bd-0367-4357-9681-9c39d91ad640
Title: demo course
Description: demo course demo
Difficulty Level: Beginner
Created: 2025-11-07T13:32:59.255Z

basic of C++ one shot
Course ID: 0c82f48f-b723-4762-8d2a-f7838ec834d2
Title: basic of C++ one shot
Description: basic of C++
Difficulty Level: Beginner
Created: 2025-11-06T17:39:40.423Z

ruby one shot lecture
Course ID: 70b6edb3-0b2e-45c7-ae87-a20ea3d183da
Title: ruby one shot lecture
Description: learn ruby in one go

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10:28 75%

Enrolled Courses

No enrolled courses yet

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10:28 75%

Create Course

Course Title
Enter your course title
Give your course a clear and descriptive title

Brief Description
Describe what your course will cover...
Provide a brief overview of the course content and objectives

Number of Units
Select unit count
Choose how many units your course should have

Difficulty Level
Select difficulty
Set the appropriate difficulty level for your target audience

Include YouTube Videos?

Generate Outline

10:28 75%

Profile

Krish
krish@gmail.com

AI Provider Settings
Choose which AI provider to use for generating course content

Outline Generation
Select provider for course outlines
Groq

Content Generation
Select provider for course content
Groq

Save Settings

Log Out

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10:29 75%

Create Course

Course Title
gemini api call learn
Give your course a clear and descriptive title

Brief Description
how to use gemini api kets
Provide a brief overview of the course content and objectives

Number of Units
4 Units
Choose how many units your course should have

Difficulty Level
Beginner
Set the appropriate difficulty level for your target audience

Include YouTube Videos?

Generate Outline

10:29 75%

Review Course Outline

Edit

gemini api key calls
Beginner

Unit 1: Introduction to Gemini API
- What is Gemini API
- Creating a Gemini API Key
- Understanding API Key Permissions
- Basic API Request Structure
- Common API Errors and Solutions

Unit 2: Making API Calls with Gemini

Generate Course Content

10:30 75%

Course Content

Generating: Course

Preparing batch generation...

10:30 75%

What is Gemini API

What is Gemini API

1 min read

Understanding the Gemini API Overview

Why this matters:

Knowing what the Gemini API does provides the foundation for any integration. It explains the services offered, the data flow, and how developers can extend Gemini's capabilities within their own applications.

Core Concepts:

- **Service Endpoint:**
  The Gemini API exposes HTTP endpoints that allow external programs to request data or trigger actions provided by the Gemini platform.
- **RESTful Design:**
  Requests follow standard REST conventions (GET, POST, PUT, DELETE) making it easy to work with common HTTP tools and libraries.
- **Authentication Model:**
  All calls require an API key, which identifies the caller and governs access to resources.

Examples:

1. **[Analogy]:**
   Think of the Gemini API as a restaurant menu: the menu (API documentation) lis dishes (endpoints) you can order, and the waiter (HTTP request) brings the dish to

10:30 75%

What is Gemini API

What is Gemini API

1 min read

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Unit 1: Introduction to Gemini API

Unit 2: Making API Calls with Gemini

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Introduction to the Course

Introduction to the Course

1 min read

Welcome to Demo Course: Overview and Objectives

Why this matters:

Introduction to the course sets the stage for the learning journey, outlining what learners can expect to achieve and the key concepts they will explore. It provides an essential roadmap for the course, ensuring learners are prepared and motivated.

Core Concepts:

- **Course Overview:**
  A brief introduction to the course, including its main topics, goals, and the skills learners will acquire by the end of the course.
- **Learning Objectives:**
  Clearly defined learning objectives that outline what learners are expected to know and be able to do after completing the course, providing a clear direction for their learni
- **Target Audience:**

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Conclusion

Here's a **professional, funding-ready conclusion** written in the same format but tailored for **your AI CourseGen app** — focusing on its unique features, motivation, and impact:

Conclusion

• A personalized AI-driven course generation platform! That's an innovative and forward-thinking project. To conclude, **AI CourseGen** is an intelligent learning platform that empowers users to create fully personalized learning paths powered by multiple AI models.

It allows users to:

- Create customized courses by simply entering a topic or idea.
- Generate structured outlines and learning modules instantly using AI.
- Edit, refine, and save outlines with flexibility before generating full course content.
- Choose between multiple AI providers (Groq, Cerebras, Gemini) for generation quality and speed.
- Revisit and access previously generated courses anytime.

For administrators, it provides tools to:

- Manage and moderate all user-generated courses.
- Add or remove AI providers dynamically.
- Oversee the overall quality and flow of content generation.

The benefits of **AI CourseGen** include:

- Breaking the barrier of procrastination and unstructured online learning.
- Helping users who struggle to learn from scattered internet content or poorly structured AI responses.
- Providing organized, step-by-step, and AI-generated learning experiences without requiring prompt engineering knowledge.
- Offering flexibility, personalization, and scalability — all in one application.
- Delivering real-time, multi-model integration for optimized learning outcomes.

By developing a user-friendly, scalable, and truly personalized AI learning application, **AI CourseGen** has the potential to redefine self-learning — transforming how individuals engage with knowledge, and making structured education accessible to everyone, instantly.

Future Scope

Future Scope

The application has immense potential for future expansion, innovation, and scalability. With continuous advancements in AI, natural language processing, and personalized learning technologies, the system can evolve into a complete intelligent education platform. The following points outline the future scope:

1. **Adaptive Learning Pathways**
   The system can integrate adaptive learning algorithms to analyze user progress and automatically adjust course difficulty, content depth, and delivery style based on each learner's behavior and comprehension level.
2. **Voice-Based Learning and Interaction**
   Future versions could integrate voice assistants or speech recognition, enabling learners to interact with AI through natural conversation—asking questions, summarizing topics, or generating quizzes via voice.
3. **Real-Time Collaboration and Peer Learning**
   Introducing discussion boards, collaborative study rooms, and mentor-student sessions can enhance engagement and provide a more community-driven learning experience.
4. **AI-Driven Assessment and Feedback**
   Incorporate automated quiz and test generation, progress analytics, and instant personalized feedback for learners to evaluate their understanding and track performance.
5. **Offline Access and Cross-Platform Synchronization**
   Implement offline content storage and seamless synchronization across devices so that learners can access their courses anytime, anywhere.
6. **Integration of Multiple AI Providers and Model Selection**
   Expand support for additional AI model providers beyond Groq, Cerebras, and Gemini—allowing dynamic model switching, benchmarking, and model-specific content generation optimization.
7. **Gamified Learning Experience**
   Add gamification features such as badges, leaderboards, and achievement rewards to make learning more interactive and motivating.
8. **Multilingual Course Generation**
   Enable AI-driven translation and localization so users can generate and learn courses in their preferred languages, broadening global accessibility.
9. **Educator and Institution Portals**
   Allow educators or organizations to upload structured content, create custom AI-powered course outlines, and share or monetize their learning paths.
10. **AI-Generated Video and Visual Lessons**
    Future versions can integrate multimodal AI (text-to-video, text-to-image) for automatically generating visual or animated explanations for subtopics, turning textual learning into immersive multimedia courses.

References

Web Resources

The following website were referred during the analysis and execution phase of the project

(1) https://developer.android.com/courses/android-basics-kotlin

(2) https://www.cheezycode.com/2021/01/kotlin-development-environment-setup.html

(3) https://developer.android.com/courses

(4) https://www.kodeco.com/books/kotlin-multiplatform-by-tutorials

Book Resources

The following books were referred during the analysis and execution phase of the project

- "Android Programming with Kotlin for Beginners" by John Horton
- "Android App development for Dummies" by Michael Burton
- "Android SQLite Essentials" by Sunny Kumar Aditya and Vikash Kumar Karn